Archaeology of Ancient Crete – the Minoans

About 4000 years ago a Bronze Age civilisation on Crete built great palaces with light wells, drains and frescoes, traded across the whole eastern Mediterranean and wrote in a script that no one today can read. We call them the Minoans, after the legendary King Minos. The Arts/Science course at Sussex was meant to give science students a subject from the humanities – and mine was the archaeology of Crete. The page has a map of Crete and an interactive figure on radiocarbon dating, where physics and archaeology meet.

> University of Sussex, Arts/Science Requirement (1975–78): Archaeology of Ancient Crete – the Minoans. The page describes the subject with the knowledge we have today; it is not a reproduction of the original course.

Arts/Science at Sussex

Sussex was founded in 1961 with the idea of "redrawing the map of learning": students were not only to learn their own subject but to see it in a wider context. Science students therefore had to take a course from the humanities or social sciences, and vice versa – the requirement the syllabus calls the Arts/Science Requirement. For a physics student archaeology was a good choice: it is a historical science built on scientific methods – from radiocarbon and tree rings to geology and DNA (see also physics in history and society).

Evans and Knossos

The British archaeologist Arthur Evans began excavating a mound at Knossos, south of Heraklion, in March 1900. He had been looking for a prehistoric script on small seal stones, and he found a huge palace with hundreds of rooms, staircases, light wells and frescoes – and thousands of inscribed clay tablets. Evans named the civilisation *Minoan* after King Minos of the Greek myth of the labyrinth and the Minotaur, and he divided its history into Early, Middle and Late Minoan periods on the basis of pottery styles. He rebuilt parts of the palace in reinforced concrete and had painters "restore" frescoes from small fragments – a project that still shapes what tourists see, and is still debated.

The palaces

The first great palaces were built around 1900 BC at Knossos, Phaistos and Malia and were destroyed around 1700 BC, probably by earthquakes – Crete lies where the African plate dives beneath the Aegean (see earthquakes). They were rebuilt even larger. A palace was at once

residence, cult centre, workshop and storehouse: in the magazines of Knossos stood rows of man-high clay jars (pithoi) for oil, wine and grain, and the accounts were kept on clay tablets. The building technique was advanced: light wells that let daylight down through several storeys, terracotta water pipes and a drainage system not surpassed in Europe until Roman times. Strangely, the palaces had almost no defensive walls.

Map of Crete



Art and daily life

Minoan art is lively and full of nature: dolphins, octopuses, lilies and birds on frescoes and pottery. The fine, thin-walled Kamares ware of the first palaces and the later “Marine Style”, with octopuses spreading their arms over the whole jar, are among the most beautiful of antiquity. The most famous fresco shows bull-leaping: young men and women seizing a charging bull by the horns and somersaulting over its back. Whether it was sport, ritual or both, we do not know. The Minoans made offerings in sacred caves and on peak sanctuaries, and female figures play a large part in their images – which is why many have spoken of a mother goddess. Some of the museums’ best-known figures, among them several ivory “snake goddesses”, are now suspected of being forgeries from the 1920s.

Linear A and Linear B

The Minoans first used a pictographic script, Cretan hieroglyphs, and then *Linear A* – a syllabary of about 90 signs. Linear A has still not been deciphered: we can pronounce many of the signs,

because they resemble those of the later Linear B, but the language is unknown. The Phaistos Disc, found in 1908, has 241 signs stamped in a spiral on both faces and is not deciphered either. *Linear B*, of which Evans found thousands of tablets at Knossos, was deciphered in 1952 by the British architect Michael Ventris, who with John Chadwick showed that the language was an early form of Greek. The tablets are accounts: sheep, wool, oil, helmets and chariots. This means that in its last period Knossos was ruled by Greeks from the mainland – the Mycenaeans.

The Thera eruption

At the height of the Minoan era the volcano on the island of Thera (Santorini) exploded in one of the largest eruptions of the last 10,000 years. The centre of the island collapsed, leaving the caldera that ships sail into today. The town of Akrotiri on Thera was buried under metres of pumice and ash, like Pompeii – but its inhabitants had fled. Spyridon Marinatos began excavating it in 1967 and found two-storey houses with intact frescoes. As early as 1939 Marinatos had proposed that the eruption and a tsunami destroyed the Minoan civilisation, and some have linked it to Plato's story of Atlantis. Today we know that the Minoan palaces went on flourishing for several generations after the eruption.

When did it happen? The archaeologists' chronology, based on Egyptian king lists and traded goods, pointed to about 1500 BC. Radiocarbon dates – among them from an olive branch buried alive in the ash – pointed to the late seventeenth century BC. The disagreement has lasted for decades; newer measurements with tree rings and the revised calibration curve of 2020 point to somewhere between about 1600 and 1550 BC. It is one of the best examples of how physics and archaeology can disagree – and grow wiser from it.

Decline and the Mycenaeans

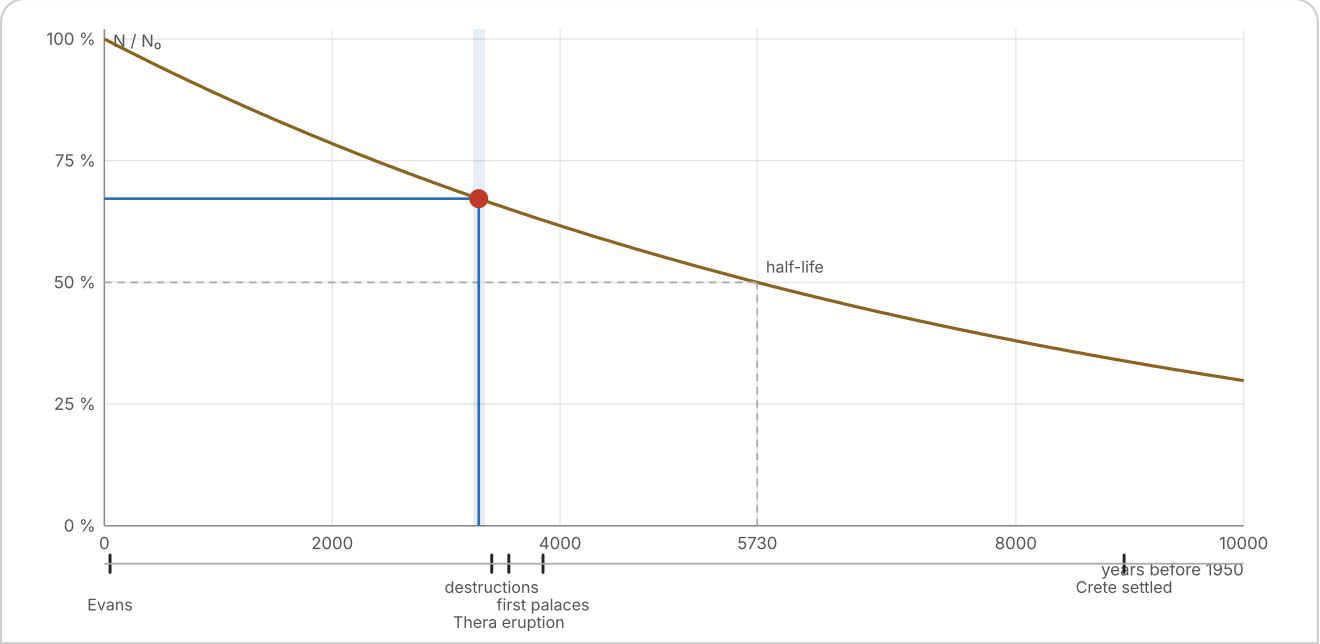
Around 1450 BC nearly all the Minoan palaces and villas were destroyed by fire. Only Knossos continued – now under Mycenaean rule, with Linear B accounts in Greek and weapons in the graves. Why it happened, we do not know for certain: earthquakes, revolt, a weakening after the Thera eruption or a Mycenaean conquest have all been proposed. By about 1100 BC, when the whole Bronze Age world of the eastern Mediterranean collapsed, Minoan culture had vanished. But its memory lived on in the myths of the labyrinth, the Minotaur, Daedalus and Icarus.

Science in archaeology

Willard Libby developed radiocarbon dating in 1949 and received the Nobel Prize in Chemistry in 1960. Cosmic rays produce carbon-14 in the atmosphere ($^{14}\mathrm{N} + n \rightarrow {}^{14}\mathrm{C} + p$), and all living organisms take it up in the same ratio to carbon-12 as the atmosphere has. When the organism dies, the uptake stops, and carbon-14 decays with a half-life of 5730 years (see nuclear physics):

$$\frac{N}{N_0} = e^{-\lambda t} = \left(\tfrac{1}{2}\right)^{t/T_{1/2}}, \qquad t = -\frac{T_{1/2}}{\ln 2} \ln \frac{N}{N_0}$$

But the atmosphere's carbon-14 content has not been constant – it varies with the Sun's activity and the Earth's magnetic field. A measured "radiocarbon age" must therefore be calibrated against wood whose age is known exactly by counting tree rings (dendrochronology). The calibration curve has flat plateaus where many calendar years give almost the same carbon-14 content – and right around the Thera eruption there is such a plateau. That is one reason the dating has been so difficult.



Other scientific methods add to the picture: *dendrochronology* gives calendar years directly; *ice cores* from Greenland show the sulphuric acid from large eruptions as layers in the ice; *thermoluminescence* dates fired pottery; *X-ray fluorescence* and *isotope analyses* show where the raw materials for bronze and pottery came from. And since the 2010s ancient DNA has transformed archaeology: a study in *Nature* in 2017 (Lazaridis et al.) showed that the Minoans and Mycenaeans were closely related and descended mostly from farmers who had migrated from Anatolia in the Neolithic.

Since 1978

- **The Thera date** has been refined with tree rings, ice cores and the new calibration curve IntCal20 – and the debate is not over.
- **Ancient DNA** has answered questions that were pure guesswork in the 1970s: where the Minoans came from, and how they were related to the Mycenaeans.
- **New excavations** – for example the palace at Galatas and the town at Petras – and underwater archaeology have widened the picture beyond the great palaces.
- **Linear A** is still undeciphered; new finds and statistical and computer-based analyses have not yet cracked the code.

- **World Heritage:** In July 2025 six Minoan palatial centres – among them Knossos, Phaistos, Malia and Zakros – were inscribed on UNESCO's World Heritage List.

Further reading

- J. Lesley Fitton: *Minoans* (British Museum Press 2002)
- Cynthia W. Shelmerdine (ed.): *The Cambridge Companion to the Aegean Bronze Age* (Cambridge 2008)
- John Chadwick: *The Decipherment of Linear B* (Cambridge 1958, 2nd ed. 1967)
- Iosif Lazaridis et al.: "Genetic origins of the Minoans and Mycenaeans", *Nature* 548 (2017) 214–218

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